



Full Length Article

Relationships between cultural orientations, phishing victimization, and phishing recognition: A cross-cultural experiment

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ABSTRACT

Background: Humans remain a critical vulnerability in the cybersecurity chain. While research has explored various behavioral factors influencing phishing susceptibility, the role of national culture and individual cultural orientations remains under-researched, representing a significant gap in the literature.

Aims: This study investigates the impact of individual cultural orientations on phishing victimization, while taking into account other relevant factors, such as self-control, risk-taking, technical training, email management practices, demographics (age and gender), and country-level economic and ICT development.

Methods: Data were collected via an online survey of university students (N = 2143) across 12 countries in Asia, Africa, North America, and Europe. Outcomes measures included phishing victimization, phishing recognition, and legitimate email recognition; the last two measures were assessed via scenarios. Data were analyzed using Signal Detection Theory and mixed modelling.

Results: Phishing victimization was significantly associated with low self-control, high risk-taking, high exposure, and poorer recognition of legitimate emails. Conversely, cultural orientations, religiosity, and country of origin had minimal effects. While phishing recognition was unrelated to victimization, the ability to recognize legitimate emails reduced victimization risk. For culturally diverse organizations, these findings suggest that cultural factors may be less critical to phishing victimization than has been previously assumed. Training users and improved self-control techniques may help protect against phishing victimization.

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1. Introduction

Online crime has become one of the most pervasive global threats in the digital age, with the number of victims and financial losses continuing to rise worldwide. In the United States alone, losses reported to the Internet Crime Complaint Center (IC3) increased from \$4.2 billion in 2020 to \$16.6 billion in 2024 (IC3, 2024). Similar upward trends have been documented across Europe (Lisiak-Felicka, 2023), West Africa (Button et al., 2025), and Vietnam (Gohwong, 2017). Among the various forms of online crime, phishing remains the most prevalent and damaging, serving as a primary driver of incidents and data breaches (Longtchi, Rodriguez, Al-Shawaf, Atyabi, & Xu, 2024; Salahdine & Kaabouch, 2019; Verizon, 2024). Indeed, phishing attacks have escalated steadily since 2016, reflecting both their adaptability and the enduring challenges of user vulnerability (Anti-Phishing Working Group, 2024).

Research shows that the human factor is crucial in phishing victimization (Kaur, Eeten, Janssen, & Fiebig, 2021; Safa, Von Solms, & Fletcher, 2016). Extant research has investigated the factors that influence users' behavior, such as personal characteristics including Information and Communications Technology (ICT) knowledge, self-control or privacy orientations, and ways in which users deal with their email (Aleroud & Zhou, 2017; Jansson & von Solms, 2011; Longtchi et al., 2024). Conversely, only a few studies have examined societal aspects such as the role of culture and proposed that cultural factors may be an important determinant of security-relevant behavior (Rocha Flores, Holm, Nohlberg, & Ekstedt, 2015).

While currently understudied, there are several reasons to explore the impact of culture on online victimization. For instance, user behavior is not easily influenced to become more careful (Bada, Sasse, & Nurse, 2015; Bullée & Junger, 2020), and therefore a good understanding of culturally influenced risk factors may increase the options for prevention. Additionally, the increasing digitalization of the economy promotes the globalization and internationalization of the workforce (Liao, Loures, Deschamps, Brezinskia, & Venâncio, 2018; Schwab, 2017; Xu, David, & Kim, 2018). According to Bonner and Wiggin (2011) 'anything that can be digitized can be globalized'. Bonner and Wiggin (2011, p. 223) noted two trends that make up this globalization trend. First, companies expand to other countries or merge internationally. Second, both blue-collar and white-collar workers are looking for work outside their country of birth. The result is a 'continuous displacement of labor' (Wadley, 2021). Cultural differences may therefore become increasingly relevant for organizations in myriad ways.

While culture serves as the primary focus, prior research suggests that other domains offer complementary explanations for why users fall for phishing attempts. For instance, the self-control model posits that individuals require sufficient self-regulation to resist the impulse to manage emails hastily and to avoid risky behaviors. Similarly, research on knowledge emphasizes that users often become victims due to their limited ability to identify malicious links. Other studies have highlighted the role of exposure to online risks and the influence of specific email management techniques, which may both stem from self-control and exert independent effects on phishing victimization. As such, the present study aims to understand how culture influences phishing victimization and email recognition, with particular attention to factors that have previously been linked to these outcomes. Specifically, we examine five domains anticipated to shape phishing susceptibility: (1) cultural orientations as reflections of national culture, (2) self-control, (3) knowledge, (4) other victimization, and (5) email management habits. We control for socio-economic factors, namely age and gender, and for national economic and ICT development.

1.1. Cultural orientations, religion, and phishing victimization

National culture. Hofstede (2011, 2024) defined culture as a set of

customs, traditions and values of a society or community, such as an ethnic group or nation. 'Culture' generally refers to a national culture, as it differs between countries (Hofstede, 2024; House, Javidan, Hanges, & Dorfman, 2002). Hofstede (2024) identified six dimensions that characterize national culture.⁴ To the best of our knowledge, no previous research has investigated the associations between cultural orientations and phishing victimization. However, several studies examined security indicators in cross-national research and analyzed data at the country level, assuming that cross-national differences reflect cultural differences, in line with (Hofstede, 2024). In order to distill plausible hypotheses, we used the findings of these cross-national studies and translated them into expectations for the present study at an individual level.

- 1) *Power distance (PDI)* is the extent to which the less powerful members of organizations and institutions (such as the family) accept and expect that power is unequally distributed. A low power distance culture values the equal treatment of everyone. In a context with a high PDI, people do what they are told to do (Hofstede, 2011). In a cross-national study of 94 countries Jeong, Grobler, Chamikara, and Rudolph (2019) found that in higher PDI countries password strength was lower (see also Bullée, Montoya, Junger, & Hartel, 2017). Therefore, we expect that a high PDI might increase the likelihood of phishing victimization and reduce email recognition accuracy.
- 2) *Uncertainty avoidance (UA)* reflects the extent to which members of a society feel uncomfortable with ambiguity and uncertainty. It captures the degree of anxiety or distrust individuals experience when faced with the unknown, as well as their preference for structure, established routines, and predictable patterns of behavior. Jeong et al. (2019) found no relationship between a country's UA and security-relevant behavior such as password strength. Accordingly, we predict that there will be no link between UA and phishing victimization.
- 3) *Individualism versus collectivism (IDV)* is the degree to which people feel independent, as opposed to favoring interdependence as members of larger wholes. Individualism means that individual choices and decisions are expected. In countries with a relatively high individualism password strength is higher (Jeong et al., 2019). We hypothesize that individualism decreases phishing victimization.
- 4) *Long-term orientation (LTO)*. In a long-term orientation, the most important work values are learning, honesty, adaptability, responsibility, and self-discipline. For example, a focus on profit in ten years' time. Research found that, in countries that score high on long-term orientation, users have more secure username and password combinations (Jeong et al., 2019). We therefore expect that a long-term orientation will lead to more precautionary behavior and therefore to less phishing victimization. Below (section 1.3) we argue that LTO is related to the self-control perspective.
- 5) *Masculinity (MAS)*. Masculinity reflects the degree to which assertiveness, competition, and toughness are socially valued within a society. In nations with cultures characterized by high masculinity, traditional gender roles are emphasized, and men are expected to display strength and decisiveness. Prior research has provided limited evidence linking a country's masculinity level to safety practices or security-related behaviors (Jeong et al., 2019; Onumo, Cullen, & Ullah-Awan, 2017). Accordingly, we do not anticipate a significant association between MAS and phishing victimization.

In the conceptualizations of a national culture, however defined,

⁴ A sixth dimension in Hofstede (2024)'s model is indulgence, referring to the degree to which societal norms allow citizens to fulfill their human desires. We did not include this dimension in the study for practical reasons, and it seemed less relevant for the present research question.

culture is a characteristic of a collective (Hofstede, 2011; Leidner & Kayworth, 2006). This approach assumes that a person's cultural orientation is determined by their nationality (Taras, Steel, & Kirkman, 2016; Vasalou, Joinson, & Courvoisier, 2010). Treating all individuals in a community or a country as identical is problematic. Taras et al. (2016) showed that about 16–21 % of the variance in cultural values resides *between* countries, showing that country is an imperfect proxy for an individual's culture. Consequently, researchers proposed to measure culture at the individual level (Sharma, 2010; Yoo, Donthu, & Lenartowicz, 2011, p. 194), and the present study uses the individual approach and measures 'cultural orientations' as proposed by Yoo et al. (2011) who provided a 'translation' of cultural values at the societal level into cultural orientations at an individual level.

Religiosity. Beyond cultural orientations, religion constitutes a central element of the social environment that can shape individual behavior. Although each faith's traditions are distinct, most religions share core values that align with broader societal norms (Johnson, Hill, & Cohen, 2011). Religious teachings commonly emphasize restraint, discipline, and self-control (Briki et al., 2015; Klanjšek, Vazsonyi, & Trejos-Castillo, 2012; Minton, 2018). Consistent with this perspective, prior research has shown that higher religiosity is associated with lower tendencies toward risk-taking (Jiang, Jiang, Kim, & Zhang, 2015; Koletić et al., 2021; León & Pfeifer, 2017). Therefore, we expect that individuals with stronger religious commitment will exhibit lower rates of phishing victimization.

1.2. The knowledge perspective

The knowledge-based perspective contends that an individual's grasp of online security concepts is a fundamental determinant of their susceptibility to phishing. The possession of technical and situational knowledge is a prerequisite for safely navigating digital environments. Conversely, a deficit in this knowledge - specifically regarding phishing techniques, email authentication, and fraud indicators - compromises a user's ability to recognize and evade malicious messages, consequently elevating their vulnerability to victimization.

Technical background. Knowledge of ICT generally reduces the likelihood of phishing victimization (Baki & Verma, 2022; Jampen, Gür, Sutter, & Tellenbach, 2020). Participants with high computer expertise are more likely to correctly identify a phishing website (Gopavaram et al., 2021; Wright & Marett, 2010). We thus expect that more IT knowledge will reduce the risk of victimization from phishing attacks.

Privacy orientation. A strong orientation toward privacy is expected to motivate more cautious online behavior and the implementation of preventive security measures, and therefore be less likely to fall victim to cybercrime (Harrison, Vishwanath, Ng, & Rao, 2015; Joinson, Reips, Buchanan, & Schofield, 2010). We therefore expect that individuals with high privacy concerns will demonstrate lower rates of phishing victimization.

1.3. The self-control perspective

Duckworth and colleagues' research (Borghans, Duckworth, Heckman, & Ter Weel, 2008; Duckworth & Kern, 2011; Duckworth, Milkman, & Laibson, 2018) proposed that self-control comprises four key dimensions: urgency, lack of perseverance, lack of planning, and, to a lesser extent, risk-taking (see also Nigg (2017)). Numerous studies have demonstrated that low self-control is associated with a wide range of adverse outcomes including accidents, poor health, deviant behavior, and victimization (Gottfredson, 2021; Hirschi & Gottfredson, 2018; Pratt, Holtfreter, & Reisig, 2010). Building on this body of work, we propose that the three concepts discussed below represent to varying degrees, facets of self-control.

Self-control. Self-control is the 'ability to wait until tomorrow for what you want today' (Logue, 1995) and to resist temptations that have negative consequences in the long run. People with low self-control take

more risks and are more impulsive, making them more likely to be both perpetrators and victims of crime (Pratt, Turanovic, Fox, & Wright, 2014). Self-control is associated with various forms of online victimization of crimes involving individual choice (Longtchi et al., 2024, p. 219; Louderback & Antonaccio, 2021) including phishing (Parsons, McCormac, Pattinson, Butavicius, & Jerram, 2013; Vishwanath, Harrison, & Ng, 2018). A high degree of self-control should protect against phishing.

Thoughtfully Reflective Decision-Making (TRDM). Where self-control is typically conceptualized as a stable personality trait, TRDM emphasizes rational and deliberate decision-making within specific situations. It involves systematically gathering information about a problem, generating and evaluating alternative courses of action, selecting the most appropriate solution, and reflecting afterward on the effectiveness of one's decision-making process (Paternoster & Pogarsky, 2009). Prior research demonstrated that individuals with higher levels of TRDM are more likely to engage in self-protective online behaviors (Howell, 2021; Howell, Maimon, Muniz, Kamar, & Berenblum, 2024) and are less susceptible to smishing (text message-based phishing) victimization (Louderback & Antonaccio, 2017). Accordingly, we expect that individuals with higher TRDM will be less likely to fall victim to phishing attacks.

Risk-taking behavior. Because low self-control predicts higher risk-taking, it seems plausible that online risk-taking in turn predicts phishing victimization. Several studies have confirmed this thesis (Koning, Junger, & Veldkamp, 2023). Abroshan, Devos, Poels, and Laermans (2021) found that a general propensity to take risks, and not necessarily a specific attitude to a specific risk, predicted link clicking and accordingly increased phishing victimization.

In the current study, risk-taking is operationalized as credential sharing. Credential sharing is relatively common, likely poses a risk for phishing victimization (Warford et al., 2022), and is a major factor in data breaches and security incidents (Verizon, 2024). We expect risk-taking will increase the likelihood of phishing victimization.

1.4. Other victimization experiences

Empirical evidence on the effects of prior cyber-victimization remains divided. Wright and Marett (2010) suggested that individuals who have been victimized become more aware of online risks and adopt safer behaviors (see also Kumaraguru et al. (2009); Kumaraguru, Sheng, Acquisti, Cranor, and Hong (2008); Mayhorn and Nyeste (2012)). In contrast, other studies have indicated that those who have experienced one form of online victimization are more likely to be victimized again (Langton, DeLiema, Brannock, & Preble, 2024; Moneva, Leukfeldt, Van De Weijer, & Miró-Llinares, 2021). Such patterns may reflect underlying common causes—specifically, low self-control and low TRDM, which can increase susceptibility to repeated victimization (van Wilsem, 2013). Additional factors, such as elevated exposure to online risks or the publication of personally identifiable information (PII) on the Dark Web, may further heighten vulnerability (Kayser, Back, & Toro-Alvarez, 2024; Snyder & Golladay, 2024).

Given these mixed findings, it is important to also consider the role of users' email management tactics. How individuals organize, prioritize, and respond to emails may both reflect and reinforce their level of self-control and awareness of digital risks—factors that can ultimately influence phishing victimization on the Dark Web (Kayser et al., 2024; Snyder & Golladay, 2024).

1.5. Email management tactics

Email management has become a highly habitualized task for many, typically executed with limited systematic analysis. This tendency toward automatic processing means that users handling a large influx of emails are likely to pay less attention to content details, resulting in a reduced capability to recognize phishing attempts (Dabbish & Kraut,

2006; Vishwanath, Herath, Chen, Wang, & Rao, 2011). Yet, the extent to which processing numerous emails—or experiencing high cognitive load online (Nasser et al., 2020)—increases susceptibility to phishing remains uncertain (Harrison et al., 2015; Sarno & Neider, 2022; Vishwanath et al., 2020). It is nevertheless plausible that more deliberate and mindful email handling provides some protection against phishing.

In addition to management techniques, exposure also plays a role: several studies have emphasized that a larger attack surface increases vulnerability to online victimization (Edwards, Larson, Green, Rashid, & Baron, 2017; Tajalizadehkhoob, Gañán, Noroozian, & Eeten, 2017). Accordingly, having multiple email accounts or managing high volumes of correspondence may not only strain attention but also heighten exposure to phishing attacks.

1.6. Socio-demographic variables and economic factors

Age. Empirical findings on age and phishing victimization are inconsistent. Most studies have found that younger individuals are at greater risk of phishing victimization (Baki & Verma, 2022; Jampen et al., 2020), although some have reported no significant effect of age (Leukfeldt, 2014) or mixed findings (Li et al., 2020). Given that the age range in the present study is relatively limited, we do not anticipate a significant relationship between age and phishing victimization.

Gender. Similar to age, the evidence regarding gender differences in phishing susceptibility is inconsistent (Jampen et al., 2020). Some studies found no gender effects (Gopavaram et al., 2021; Holtfreter, Reisig, & Pratt, 2008), others found that men perform slightly better than women in detecting phishing messages (Bullée & Junger, 2020; Sheng, Holbrook, Kumaraguru, Cranor, & Downs, 2010) and some studies stated that women may display greater caution and risk aversion, leading to lower susceptibility (Holt & Turner, 2012). These discrepancies may be attributable to differential exposure rather than inherent susceptibility (Koning et al., 2023). Given these mixed findings and limited theoretical clarity, we do not expect a significant relationship between gender and phishing victimization in the present study.

Economic development and ICT development level. Prior research has indicated that countries with higher levels of economic prosperity and more advanced ICT infrastructures tend to exhibit higher rates of cybercrime victimization (S. Chen et al., 2023; Kigerl, 2021; Srivastava, Das, Udo, & Bagchi, 2020). This pattern may reflect both greater online activity and broader access to digital technologies, which increase exposure to potential threats. Accordingly, we expect that economic and technological development will shape patterns of cybercrime victimization and therefore include these variables as controls in the present analysis.

1.7. Current study

The present study examines how cultural orientations influence phishing victimization and email recognition, with particular attention to individual and contextual factors previously linked to online security behavior. Specifically, we investigate five domains that are expected to shape susceptibility to phishing: (1) cultural orientations as reflections of national culture, (2) self-control, (3) knowledge, (4) other victimization experiences, and (5) email management habits. Together, these domains capture both stable dispositional factors and situational behaviors that may jointly determine vulnerability to phishing attacks. In addition to these focal variables, we control for socio-demographic characteristics (age and gender), and for macro-level indicators of national economic and ICT development, as both factors have been shown to influence patterns of cybercrime victimization. Accordingly, this study seeks to address two central research questions:

1. How are individual cultural orientations related to phishing victimization?

2. How do other risk factors—such as self-control, knowledge, other victimization, and email management habits—contribute to phishing victimization beyond the influence of culture?

By integrating psychological, cultural, and behavioral perspectives, this study aims to provide a more comprehensive understanding of why some individuals are more vulnerable to phishing than others across diverse cultural contexts.

2. Method

2.1. Participants

This study utilized a multinational student sample to balance key methodological considerations. To ensure meaningful variability in cultural orientations while controlling for extraneous demographic factors like age and socioeconomic status, we recruited participants from universities across four continents. This approach maximized cultural diversity while maintaining demographic comparability. Reasons for focusing on young users include: young people spend relatively more time online (ITU, 2024); and they are more likely to be exposed to phishing attempts (Koning et al., 2023; Office for National Statistics, 2022). Also important is that university students represent the future workforce, which is increasingly globalized, mobile and digitally active (Bonner & Wiggin, 2011; ITU, 2024). This makes them an interesting group for security research. Recruiting participants through universities across four continents allowed us to examine phishing-related outcomes in a young and digitally active population while limiting confounding by age, socio-economic factors, and educational context. At the same time, there may be disadvantages to this approach, which we also acknowledge. University recruitment—particularly via a – in most locations - English-language questionnaire—may include internationally mobile students and individuals with cross-cultural exposure. Such immersion in globally shared ICT environments may reduce between-country cultural contrasts and attenuate associations between cultural orientations and phishing outcomes. Accordingly, findings related to culture are interpreted as reflecting individual-level cultural orientations within a globally connected student population. We further discuss the advantages and disadvantages of student sample in the Discussion (section 4.3).

Participants were recruited by the authors at their respective institutions, resulting in a sample from 14 universities in 12 countries across four continents.⁵ All participants completed an online Qualtrics questionnaire hosted by the University of Twente. The sample consisted predominantly of students. In Poland, participants were drawn from the Police Academy in Szczytno. Although this institution trains future law enforcement professionals, its students were treated as comparable to university students due to the academy's academic status; only a small number of participants may have been active police officers. A special case is the Swiss subsample, which comprised academic staff from ETH Zürich recruited through the university's Risk Centre newsletter.

To capture variation in ICT-related knowledge, the sample included students from diverse academic disciplines, including Computer Science, Science, Technology, Engineering, and Mathematics (STEM), as well as the Social Sciences. We primarily recruited first- and second-year students to mitigate potential survey-response bias, as prior research indicates that extensive survey experience can alter response patterns (Stocké & Langfeldt, 2004). The questionnaire was administered in English, except for the Polish version, which was translated by the authors (PO and RP).

⁵ 35 Indian students who followed an exchange program with the University of Twente filled out a questionnaire while they were in the Netherlands. In the Appendix, Table 1, we did not count them as a separate country or university. If we were to do that, the sample would consist of students from 14 universities in 12 countries.

In total, 2790 participants began the questionnaire and 2193 completed it. After excluding cases with missing data, the final analytical sample comprised 2143 participants, representing 77.8% of the initial sample. Completion rates varied by university (see Appendix A below, Table 1). In some locations, students received compensation in the form of course credit or monetary incentives. Descriptive statistics for all key variables are provided below, in Appendix A below (Table 2).

2.2. Procedure and concepts

The online survey consisted of a phishing recognition test and several background questions, including validated scales for different (cultural) concepts. The complete questionnaire is available online (Appendix B).

2.2.1. Outcome measures

There are several ways to measure phishing (Buil-Gil, Trajtenberg, & Aebi, 2025; Distler et al., 2023). The present study combines two common methods: 1) asking about victimization experiences and 2) scenarios or (real) email classification tasks.

The use of victimization surveys, a methodology originating in criminology, is the established standard for cross-national crime comparison. This is because they circumvent the confounding effects of differing legal systems and reporting practices across countries (Aebi, Killias, & Tavares, 2002; Buil-Gil et al., 2025; Dijk, 2007; Messner & Zimmerman, 2014). Despite known limitations, these surveys have been foundational to the field since their introduction in the 1970s (Aebi et al., 2002; Buil-Gil et al., 2025; Messner & Zimmerman, 2014; National Academies of Sciences, Engineering, & Medicine, 2018; Rand, 2007).

In the present study, participants were asked: *'In the past 12 months, have you provided your login credentials (e.g., username and password) or other sensitive information (e.g., credit card details) as a result of phishing via an email or a website?'* In line with Furnell's (2007) recommendation, we operationalized phishing victimization with an explicit definition to mitigate conceptual ambiguity and standardize participants' interpretation of the key construct (Furnell, 2007).

Scenarios or email classification tasks, circumvent issues of memory decay by measuring 'pure phishing susceptibility' independent of users' email management tactics and opportunity factors. In these tasks, participants are presented with a series of emails and are required to categorize each one, typically by judging its legitimacy (e.g., phishing or legitimate) and/or stating their intended action (Anandpara, Dingman, Jakobsson, Liu, & Roinestad, 2007; Canfield, Fischhoff, & Davis, 2016; Furnell, 2007; Parsons et al., 2013; Parsons, McCormac, Pattinson, Butavicius, & Jerram, 2015).

In the scenario-based approach, participants were instructed to assume the role of a fictional persona, 'Chris Bos', and to evaluate emails in this character's inbox as if they were their own. The emails appeared to be from various organizations with which Chris Bos was presumed to have a relationship. The introduction of the emails was as follows: *'Imagine that you are Chris Bos. You have received 12 emails in your inbox. Every email seems to come from an organization you do business with. At first glance, every email seems important to you. It's your job to inspect each email and decide whether you think it's a legitimate email or a phishing email.'* Participants had to decide for each of these emails whether they thought it was a phishing email or not (question: *'Do you think this is a phishing email?'* with answer options *'Yes (this is a phishing email)'* and *'No (this is a legitimate email)'*).

All phishing emails were created for this experiment, using made-up organizations, to avoid the brand familiarity effect. To ensure cross-national comparability, the phishing emails were designed with culturally neutral, globally understandable content.

We deliberately avoided references to country-specific institutions to guarantee that all participants, irrespective of national background, could interpret the stimuli consistently. For each scenario, participants were shown the full email, including the sender's address. The stimuli varied in their requested action: six emails contained a hyperlink (with a

Table 1

Main characteristics of the 12 email messages: sending organization, topic and requested action from the participant.

#	Organization	Topic	Action requested
Legitimate e-mails			
1	art-library.com	Password change required	Click
2	XBank	Suspicious activity with account	Click
5	upSync	Storage out of space	Click
7	CloudAccount	Account locked, reply for assistance	Reply
11	New Salary Scales	Salary info in attachment	Attachment*
9	I need your help	Colleague - do you have time to help with a short request? Please reply if you are available	Reply
Phishing e-mails			
4	CryptchainVPN	Suspicious activity with account	Click
6	FL-Bank	Payment notification	Click
3	Air Storage	Storage inactivity	Click
8	Gift card	Colleague request for help	Reply
10	Review request	Review file in attachment	Attachment*
12	Thank you for your order	Shipment on the way - please review details as in attachment	Attachment*

* The email image suggested the presence of an attachment by an icon, but there was no actual attachment to download.

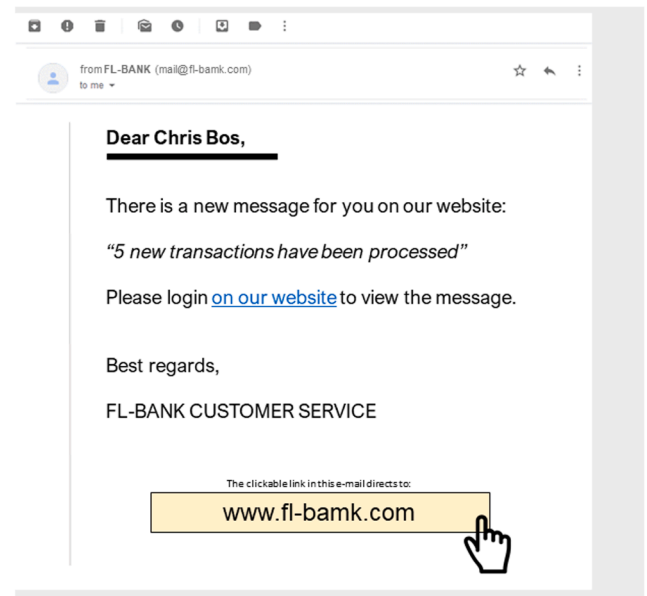


Fig. 1. Picture of phishing email # 6.

mouse-over function to reveal the destination URL), three prompted a direct reply, and three suggested a file attachment was included. For practical reasons, we only included 12 emails. The questionnaire was already relatively long, and we were concerned that adding more emails would lead to respondent dropout.

Accurate classification required participants to detect inconsistencies by cross-referencing the sender's address and any hyperlinks with the email's content. A phishing email is non genuine if the domain the URL in the mail redirects to differs from that of the organization (as provided in the task). Two outcome measures were derived from the task: (1) the number of phishing emails correctly identified and (2) the number of legitimate emails correctly recognized. The complete set of email stimuli used in the phishing recognition task is provided in the questionnaire (Appendix B online). Table 1 summarizes their main characteristics, including the order of presentation, represented organization, message topic, and requested user action. An example of one phishing email is presented in Fig. 1.

Table 2
Concepts in the study.

Phishing	Description, source if not from current authors	Categories
Score phishing	Scenario-based phishing recognition test	From 0 to 6
Score legit	Scenario-based phishing recognition test	From 0 to 6
Victim of phishing	Have you provided your login credentials (e.g., username and password) or other sensitive information (e.g., credit card details) as a result of phishing via an email or a website in the last 12 months?	No/don't know (0) or yes (1)
Cultural orientations, religion, nationality (Yoo et al., 2011)		
Power Distance	Example: 'People in higher positions should make most decisions without consulting people in lower positions' & 'People in higher positions should avoid social interaction with people in lower positions.'	High = high power distance Categories: Strongly disagree (1) Disagree (2) Neutral (3) Agree (4) Totally agree (5) High = avoiding uncertainty
Avoiding uncertainty	Example: It is important to have detailed instructions so that I always know what is expected of me' & 'Standardized working procedures are helpful.'	High = high on collectivism
Individualism vs Collectivism	Example: 'Individuals should stay with the group even when they are struggling' and 'Individuals should sacrifice self-interest for the group'	High = high long-term orientation
Long-term orientation	Example: 'Careful handling of money & personal stability and stability' & 'Working hard for success in the future'	High = high on masculinity
How important do you think next is in life?	Example: 'It is more important for men to have a professional career than for women' & 'There are jobs that a man can always do better than a woman.'	No (1) or yes (2), and 'rather not say' (0)
Masculinity	'Do you consider yourself a practicing religious person?' 'In which country did you mainly live in your youth?' This question meant to specify the country where the participant had been socialized.	USA was the reference category, separate codes were created for Germany, Netherlands, Poland, UK, India, Vietnam, Nigeria, South Africa, Canada, Mexico. Other countries were coded as 'else'.
Personality, personal characteristics		
ICT knowledge		
Technical training	The participants were asked to what extent they took ICT or technical courses. A total of 100 points had to be awarded. <i>Transformation:</i> Because this variable was skewed, we used a log transformation.	Percentage of technical courses. No/don't know (0) or yes (1)
Privacy orientation	We used the second dimension of the Privacy Orientation Scale of Baruh and Cemelcilar (2014) which measures the 'Concern about one's own informational privacy'. Example item: 'I'm worried	A higher score means a higher privacy focus.

Table 2 (continued)

Phishing	Description, source if not from current authors	Categories
Self-control	<i>that people around me know too much about me'</i>	
Self-control scale	Dysfunctional impulsivity scale, adapted from Dickman (1990), and then tested by (Pechorro et al., 2021). Average score calculated based on 12 items (each rated on a 3-point scale). Example item: 'I often say what comes into my head without thinking first' and 'Often I don't spend enough time thinking about a situation before I act.'	A higher score means more self-control.
Thoughtful Reflective Decision Making (TRDM)	TRDM is measured by 4 questions (Paternoster & Pogarsky, 2009), such as 'When you have to solve a problem, one of the first things you do is get as many facts as possible about the problem' • When you're trying to find a solution to a problem, you're usually trying to come up with as many different approaches to the problem as possible • When making decisions, you generally use a systematic method to assess and compare alternatives • After performing a solution to a problem, you usually try to analyze what went right and what went wrong	A higher score reflects more thoughtful reflective decision-making.
Risk-taking	Have you ever shared personal/online login credentials with... 1) Your partner? 2) Other family members? 3) close colleagues?	Sum of the number of times the participant mentioned sharing.
Other victimization	Consumer fraud: In the last 12 months, have you made an online purchase of a product or service that you did not receive or that was a scam?	No (0) yes (1)
Email accounts, number of emails, email tactics		
# of accounts	How many email accounts do you have that you actively use, besides your university email account?	Number
# Emails	How many emails do you receive on average per day?	In multilevel analysis (1) 12.5 (2) 37.5 (3) 75 (4) 150 (5) 250 (6) 350
Email Management Tactics	Although not explicitly created by the authors, we assumed that: Handling the flow of incoming emails is measured by the following items (1) I check my email as soon as I see or hear that a new message has arrived.	Never (-2) Rare (-1) Uncommon (0) (continued on next page)

Table 2 (continued)

Phishing	Description, source if not from current authors	Categories
		Common (1) Always (2) See item (1)
	(2) I restrict myself to checking my email at specific times of the day.	
	(3) I keep email messages in my Inbox as a reminder of things I need to do.	See item (1)
	(4) I leave email messages in the inbox after reading them.	See item (1)
	(5) Scale: Organized and Structured Email Management	Recoded as item (1)
Socio-demographic variables		
Age	Measured in whole years. To avoid overly influential outliers, the age was capped at 40, by recoding of all values above 40 to 40; this affected 39 participants.	
Gender	Woman (0), man (1), other (3).	
Economic factors		
GDP	Gross domestic product per capita in current (2023) US\$ (World Bank, 2024).	Standardized in multilevel analysis
ICT Development Index (IDI)	The ICT Development Index (Information and Communication Technologies) describes the level of technological expertise and access of countries.	

2.2.2. Explanatory factors

Following the phishing recognition test, participants were asked to complete a battery of background questionnaires measuring the constructs outlined in the introduction (see Table 2 for a complete list of predictor variables). Key measures are summarized below.

Cultural orientation. Cultural orientations were assessed using Yoo et al.'s (2011) individual-level scales, which operationalize the first five cultural dimensions defined by Hofstede (2024).⁶ We also measured nationality (i.e., the country where the participant spent most of their childhood) and religiosity, treating both as additional cultural factors that may influence victimization.

Technical training. Technical training was operationalized as the percentage of technical courses completed by each participant.

Other victimization. Other victimization was assessed by asking participants whether they had experienced consumer fraud, one of the most prevalent forms of online crime worldwide (Reep-van den Bergh & Junger, 2018; Woods & Walter, 2022).

Email accounts, email volume, and management tactics. Participants reported their number of active email accounts and average daily email volume. Email management behavior was assessed using an adapted version of Dabbish and Kraut (2006)'s Email Management Scale, in which 'work-related' items were modified to 'university-related' to suit the student sample.

A confirmatory factor analysis revealed four items loading strongly on a single factor (eigenvalue = 2.7), explaining 34.2% of the variance. These items were: I try to keep my inbox size small; I delete university-related email messages after I read them; I manually organize my email messages as soon as they come in; I organize my email messages into

separate folders. The resulting Organized and Structured Email Management (OSEM) scale was computed by averaging these items (Cronbach's $\alpha = 0.73$). For parsimony, the first item was excluded from multilevel analyses, as it was unrelated to any dependent variable.

Socio-demographic characteristics. Participants provided information on age and gender, which were included as potential control variables given their possible indirect associations with phishing victimization.

Economic and ICT development. To account for national-level contextual factors, we incorporated two indicators. Economic development was represented by Gross Domestic Product (GDP) per capita (USD, 2023) (World Bank, 2024), defined as the total annual value of goods and services produced domestically, excluding income earned abroad (Britannica Dictionary, 2024). Technological development was measured using the ICT Development Index (IDI, 2023) (ITU, 2024), which combines eleven indicators reflecting ICT access, usage, and skills.

Both GDP and IDI scores corresponded to the countries where participants were currently studying, as the phishing victimization and recognition questions referred to the preceding year. This approach ensured alignment between the data collection period and participants' current digital environments. India did not have a published IDI score for 2023 and was therefore coded as missing.

2.3. Analytical approach

We used three outcome measures for our analysis: the email recognition test of phishing, recognition of legitimate emails, and victimization of phishing. Since the data was collected in several locations, the data had a hierarchical structure with likely dependency between the participants within these groups. To account for this, we used mixed modelling (with the 'lme4' package for R (Bates, 2010)). We added a random intercept point to each model based on the data collection group (see Appendix A, Table 1 for all groups). As can be seen in Table 1 (Appendix A), some data collection groups have a relatively low number of participants. For each of the three outcome measures, two models were constructed. In a first step, null models were fitted. These are multilevel models without any explanatory variables. They indicate what proportion of the total variance in an outcome variable relates to differences between locations and what proportion to differences between individuals within locations. In a second step we added consumer victimization that, if it occurred earlier, might have raised awareness and improved the recognition of emails as phishing or legitimate; and the email recognition variables, depending on the dependent variable. In the third step, the goal was to build more parsimonious models. For this purpose, all variables that were statistically non-significant in the selected models were removed. All variables were standardized to improve the comparability of the estimates.

Finally, to analyze the responses to the 12 emails, we applied Signal Detection Theory (SDT) (Aktepe & Heck, 2026; Vuorre, 2025; Zloteanu & Vuorre, 2024). SDT offers a complementary approach to mixed-effects modeling, operating on distinct theoretical principles. As previously noted, the participants' task was to determine whether an email was phishing or legitimate. Within the SDT framework, responses to these tasks are classified into four categories: hits (H), misses (M), false alarms (FA), and correct rejections (CR). The present analysis estimates classification performance for each email based on participant responses, allowing for the calculation of two key measures. First, it provides a measure of decision sensitivity (d'), representing the participants' ability to discriminate between phishing and legitimate emails. Second, it quantifies response bias via the decision criterion (c), which establishes the decision threshold and reflects an overall tendency to classify emails as either phishing or legitimate. The value of this criterion can be influenced by external factors, such as instructions or expectations (Zloteanu & Vuorre, 2024). Our analytical model accounts for email type, individual variation among participants, the ground truth of the

⁶ As was mention in section 1.1, the sixth dimension, 'Indulgence,' was omitted for practical reasons due to questionnaire length constraints and a prioritization of other constructs.

stimuli, and the actual choices made. Consequently, the analysis utilizes a trial-level dataset of email responses rather than aggregated individual scores. Full methodological details are available in the online supplementary materials (Appendix C).

To ensure comparability between variables, data across all analyses were transformed into z-scores ($M = 0$, $SD = 1$).

2.4. Institutional review board

The study was approved first, by the University of Twente's IRB (Faculty of Behavioral, Management and Social Sciences, number 221,233). Thereafter the study was also approved by the ethics committees of the University of Nottingham (CS-2022-R33); the University of Portsmouth (Faculty of Humanities and Social Sciences; reference number FHSS 2023–018); the Nelson Mandela University Human Ethics Committee (reference number H22-ENG-ITE-EXT-002), the University of South Florida (STUDY005234), the University of Texas at El Paso (2018,942–2), the University of Nigeria, College of Medicine Research Ethics Committee (Ref. no 0160/04/2022), and Simon Fraser University (REB, #30,001,957). Other universities and schools did not require additional IRB approval, either because there was no IRB or because the initial IRB approval of the University of Twente, which listed all participating universities, was sufficient.

3. Results

The main sample characteristics are presented below, followed by the results of the multilevel analysis. Detailed information is available online (Appendix C).

3.1. Descriptive results

The final sample, as mentioned above, consisted of 2143 students. The average age of the students in the study was 22.2 (SD 4.8). The vast majority (91%) were between 17 and 27 years old, and 9.4 percent were 40 years or older. There were differences between the recruitment locations. Participants of ETH Zürich were relatively old (37.5), also relatively older were the Indian students who visited the University of Twente (26 years); the students of LMU Munich in Germany and the University of Portsmouth (UK) were slightly older: on average, they were between 24.2 and 26 years old. The country where students grew up was generally the country where they studied, but there were a few special cases. Of the Texan students, 17.8 percent grew up in Mexico, At the University of Twente, 42.8 percent of the students grew up in Germany. Both the English universities and the LMU München University (Germany) had a relatively large proportion of international students.

As expected, cultural orientations varied significantly across the country in which respondents grew up. Respondents from Vietnam (score=−0.06) placed more importance on power distance compared to respondents from other nationalities (scores ranging from −1.1 to −1.3). Respondents from South Africa and Nigeria scored relatively high on uncertainty avoidance (scores of 1.2 and 1.2 respectively); while respondents who grew up in Vietnam had the lowest scores on uncertainty avoidance (score=0.07). Respondents from Vietnam, Nigeria, South Africa and India placed the most value on collectivism, while respondents from Mexico, Poland and the Netherlands placed more value on individualism (respondents from all three scored −0.2). Respondents from Nigeria and South Africa placed more emphasis on long-term orientation (scores of 1.4 and 1.2 respectively), while Vietnamese and German respondents attached the least importance to long-term orientation (respondents from both scored 0.8). Respondents from Nigeria, South Africa, Poland and Vietnam favored masculinity (scores ranging from −0.4 to 0.1), while respondents from Mexico, Canada, the United Kingdom, the Netherlands, and Germany scored the lowest on this cultural orientation (scores ranging from −1.2 to −1). More extensive description of the sample as well as bivariate analysis are presented

online (Appendix C)

Prevalence of victimization. With regard to actual victimization, 8.3 percent of participants indicated that they had been a victim of phishing. While anyone with the information in the scenario could have provided the correct answer, the number of correct answers ranged between zero and 6, which was the maximum, as there were six phishing and six legitimate scenarios (Fig. 2). It is noteworthy that students tended to rate more emails as phishing than as legitimate. 37.4% of participants correctly identified 5 or 6 phishing emails, while only 18.8 percent correctly identified 5 or 6 legitimate emails.

The model-based SDT analysis showed that sensitivity was $d' = 0.59$, which was significantly higher than zero ($b = 0.59$, $SE = 0.267$, $z = 2.20$, $p = .028$). The z-scores of the respondents' answers ranged from −2.3 to 3. These results indicate that while participants performed significantly better than chance, their overall ability to distinguish phishing emails from legitimate ones remained limited. Although they were able to distinguish phishing from legitimate emails, the level of this ability was limited. The mean value of the criterion (c) was −0.21, which is significantly lower than zero ($p < .001$). This negative criterion indicates a liberal response bias, meaning that participants were generally more inclined to classify emails as phishing.

3.2. The multilevel analysis

The main findings regarding the full model and the parsimonious model are presented in Table 3.⁷ To facilitate interpretation, all continuous variables were transformed to z-scores. Consequently, the regression coefficients indicate the expected change in the outcome variable associated with a one-standard-deviation change in an explanatory variable. Phishing victimization is the only exception, as this variable is coded as a binary (0–1) indicator. As a rule of thumb, standardized coefficients below .10 may be considered small, whereas coefficients above .50 may be considered large (Cohen, 1988).

All coefficients in Table 3 are based on multivariate analyses. This means that the reported effects remain statistically reliable after controlling for all other variables included in the models. Although t-values were computed, they are not reported in order to limit the amount of numerical output; statistical relevance is indicated instead ($< .001$; $< .01$; $< .05$; $< .10$). Descriptive statistics for all variables in the analysis are included in appendix A (below, Table 2). The bivariate analysis and more detailed information on the multilevel model can be found online (Appendix C). The main results are summarized below, based on the model 2 findings.

Victims of phishing. Participants who were able to avoid victimization were characterized by a relatively high score on long-term cultural orientation (estimate=−.20), considered themselves non-religious (estimate=.21), did not have technical training (estimate=.23), had higher self-control (estimate=−.55), and took less risk (estimate=.48). They had a lower number of email accounts (estimate=.26), and reported that they kept their emails in their inbox as a to-do list (estimate=−.20). They were better at recognizing legitimate emails (estimate=−.19), and were less likely to be victims of consumer fraud (estimate=.50).

Phishing email recognition. Participants who scored high on recognizing phishing were relatively low on masculinity (estimate=−.07), and on taking risks (estimate=−.04), they did leave email messages in their inbox after reading (estimate=.04). They are also more likely to be male (estimate=.10), and relatively older (estimate=.09). These estimates are generally lower than in the analysis of phishing victimization.

Legitimate email recognition. Only four variables related to the recognition of legitimate emails. Compared to American participants,

⁷ We mention the estimates that are statistically significant, unless otherwise stated.

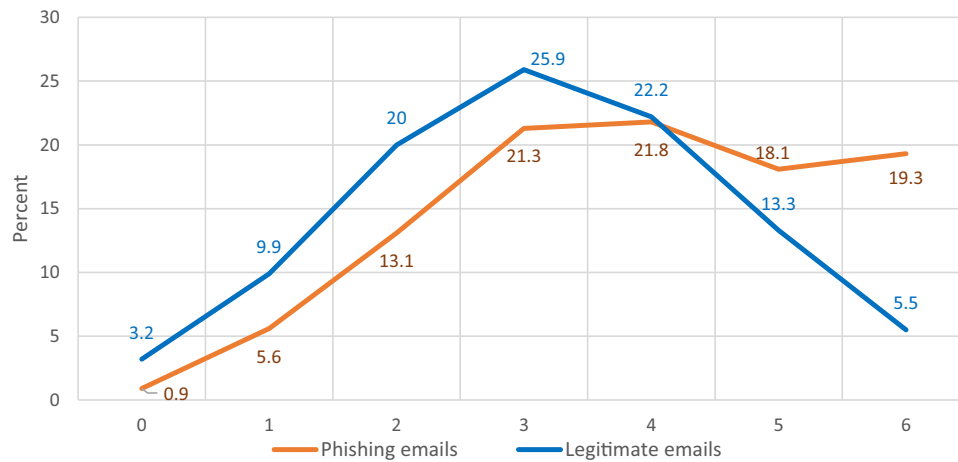


Fig. 2. Percentages of correct categorization of phishing and legitimate emails (N=2143).

Canadian participants were less able to recognize legitimate emails (estimate=-.10). In addition, participants who were good at recognizing legitimate emails had a relatively large number of email accounts (estimate=.07), but did not limit themselves to checking their email inboxes at specific times (estimate=-.09). Finally, they were relatively less likely to be victims of phishing (estimate=-.06).

The two last rows in Table 3 present the residual variance within and between locations. Overall, for all three outcomes, we find that 80–90% of total variance relates to differences within locations. To estimate the extent to which our explanatory variables explain the totality of the variance between locations one can compare the variance between locations in the null model (thus without the explanatory variables) with the variance between locations with all explanatory variables included in the model (we use model 2 data). This comparison indicates that regarding phishing victimization only a small amount of variance between locations remains. The null model shows .040 variance between locations. Of this only .07 remains in model 2. This implies that over 80% of the null model variance between locations is accounted for by the model 2 explanatory variables. With regard to phishing recognition, about half the variance between locations still remains to be explained when taking the explanatory variables into account. The null model shows .17 variance between locations of which .09 remains in model 2. With regard to recognizing legitimate emails, the variance between locations hardly changes when taking into account the explanatory variables. In the null model, the variance between locations is .07, while it equals .06 in model 2. The effect of the explanatory variables on variance at the individual level turns out to be minimal.

4. Discussion

Humans tend to conform and be relatively trustworthy by nature (Dawkins, 1993; Morgan & Laland, 2012). These same qualities, however, also render them vulnerable to social engineering tactics such as phishing (Christin, Egelman, Vidas, & Grossklags, 2011; Junger, Montoya-Morales, & Overink, 2017). Yet, the extent of this vulnerability varies, depending on factors such as cultural orientation, knowledge, self-control, and everyday digital habits. The current study examined how these individual and contextual factors influence phishing victimization, with a particular focus on cultural orientations.

In today's increasingly globalized workforce, cultural diversity is growing within organizations, making it essential to understand whether and how cultural differences affect cybersecurity behavior. Because no prior research has directly examined cultural orientations and phishing victimization at the individual-level, our expectations were informed by studies on national culture, such as those by Onumo et al. (2017) and Jeong et al. (2019).

Data were collected via an online survey administered to (mostly) university students from 14 institutions, located in 12 countries across Asia, Africa, North America, and Europe, yielding a total of 2143 participants. By incorporating a diverse, cross-cultural dataset, we addressed a major gap in cybersecurity research, which has long been dominated by participants from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies (Henrich, Heine, & Norenzayan, 2010; Kaur et al., 2021). Signal Detection Theory (SDT) was used to evaluate how effectively respondents distinguished phishing from legitimate emails and to detect any overall response bias.

This sample was relatively homogeneous in terms of socioeconomic background and educational level, allowing for meaningful cross-cultural comparisons. Moreover, university students represent the emerging global workforce, whose digital practices will shape future organizational security environments.

Overall, 8.3% of participants reported having been victims of phishing. Participants correctly identified an average of five out of six phishing emails (37.4%), but only 18.8% correctly recognized the same number of legitimate emails. Respondents were able to distinguish phishing from legitimate emails, with $d' = 0.59$, which is a modest improvement over chance. The criterion $c = -0.21$ indicates that participants were more inclined to classify messages as phishing; while this may have reduced the risk of missing a threat, it simultaneously increased the number of false alarms. This general bias toward labeling messages as phishing is possibly due to the task framing—participants were explicitly asked to evaluate whether emails were fraudulent. Similar 'subject expectancy effects' have been noted in previous scenario-based studies, where participants tend to err on the side of caution (Anandpara et al., 2007; Furnell et al., 2007; Parsons et al., 2015; Crgol, 2024).

In the following discussion, we outline the broader patterns observed in our findings. To aid interpretation, we reference relevant bivariate relationships (tables not shown), and focus on the key risk factors for phishing victimization. It is worth noting that while several independent variables were correlated with the outcomes at the bivariate level, only a subset remained significant in the multilevel analyses—potentially explaining why some associations found in earlier studies were not replicated here.

4.1. Correlates of phishing victimization

This study examined how cultural, individual, and behavioral factors shape phishing victimization and email recognition across mainly student samples from twelve countries and four continents. The findings reveal that phishing victimization is driven primarily by individual-level characteristics—most notably low self-control, risk-taking behavior, and

Tabel 3

Multi-level analysis: full and parsimonious models; estimates and statistical significance (p).

	Phishing victim				Phishing recognition				Legitimate email recognition			
	Model 1		Model 2		Model 1		Model 2		Model 1		Model 2	
	Estimate	p	Estimate	p	Estimate	p	Estimate	p	Estimate	P	Estimate	p
(Intercept)	-3.12	***	-3.07	***	0.06		0.07		0.06		0.02	
Power distance	0.03				-0.01				0			
Uncertainty avoidance	-0.02				-0.01				0.03			
Collectivism	0.14				0.03				0.01			
Long-term orientation	-0.23	**	-0.20	**	-0.01		-0.01		0.03		0.04	.
Masculinity	0.01		0.08		-0.06	*	-0.07	**	-0.03		-0.04	
Religiosity (Ref.cat.=no)	0.22	*	0.21	*	-0.02		-0.02		0.01		0.01	
Religiosity unknown	0.08				-0.01				-0.02			
Germany (Ref.cat.=USA)	-0.02				0.07	.			-0.04			
Netherlands	-0.12				0.06	.			-0.06	.		
Poland	0.09				-0.07				-0.07			
UK	-0.01		-0.05		0.09	*	0.06	.	-0.02		0.04	
India	0.06				0.04				-0.05			
Vietnam	0.31				0.09				-0.06			
Nigeria	0.17				0.07				-0.1			
South-Africa	0.11				0.02				-0.11			
Canada	0.08		0.03		0.11	.	0.06		-0.17	**	-0.10	*
Mexico	-0.07				0.01				-0.01			
Nationality else	0.06		0.01		0.02		-0.02		-0.12	**	-0.05	.
Technical training	0.19	*	0.23	**	0		0.01		-0.01		-0.02	
Privacy orientation	-0.07				-0.01				-0.01			
Self-control	-0.58	***	-0.55	***	0.03		0.04	.	0.01		0.02	
TRDM	0.06				0.02				0.03			
Risk behavior	0.47	***	0.48	***	-0.05	*	-0.04	*	0.02		0.02	
Number of email accounts	0.30	***	0.26	**	0.08	***	0.08	***	0.07	***	0.07	**
How many emails on average daily?	-0.04				0.01				-0.04			
Organized & struct. email management	0.16				0.02				-0.01			
(2) I restrict myself to check email	0.1		0.17	.	0		0		-0.09	***	-0.09	***
(4) Keep email in inbox as 'to do' list	-0.29	**	-0.20	*	0.03		0.04	.	0.04	.	0.04	
(5) I leave emails after reading	-0.05		-0.09		0.05	*	0.04	*	0.01		0.02	
Gender (Ref.cat.=female)	-0.15		-0.12		0.10	***	0.10	***	0.03		0.03	
Gender unknown	-0.11				0.03				0.04	.		
Age	-0.2		-0.14		0.08	***	0.09	***	0.02		0.02	
Age unknown	-0.19				-0.02				-0.02			
IDI	-0.11				0.14				-0.09			
GDP World bank	0.28				0.11				0.09			
Recognition phishing emails	-0.06				0		0		-0.05	*	-0.06	*
Recognition legitimate emails	-0.16	.	-0.19	*	-0.02		-0.01		0		-0.001	
Victim cons. fraud (Ref.cat.=no)	0.48	***	0.50	***								
Random effects												
Variance between locations	.00		.07		.08		.09		.11		.06	
Variance within locations	—		—		.86		.86		.92		.92	

*** < 0.001; ** < 0.01; * < 0.05; '.' < 0.1

exposure—rather than by cultural or demographic factors.

Among Hofstede's cultural dimensions, only long-term orientation emerged as a significant predictor of phishing victimization, with higher scores associated with lower susceptibility. This result aligns with Jeong et al. (2019), who found that secure online practices were more common in societies emphasizing long-term thinking. Although conceptualized as a cultural factor, long-term orientation also reflects a dispositional element akin to self-control, which may explain its predictive power. Other cultural dimensions—including individualism, masculinity, uncertainty avoidance, and power distance—showed no significant associations, suggesting that cultural influences may be less pronounced at the individual level.

Unexpectedly, participants who considered themselves religious were more likely to be victims of phishing, contrasting with previous studies showing that religious individuals engage in less risky behavior (Koletić et al., 2021; León & Pfeifer, 2017). This finding echoes Jerrim (2023, p. 19), who reported minimal cross-national differences in

phishing behavior among adolescents, suggesting that within-group psychological or behavioral factors may outweigh country-level effects. However, Jerrim (2023)'s sample is quite different from the present study. A possible explanation is that religiosity taps into aspects of culture that remained unmeasured by the Hofstede scales (Yoo et al., 2011). Another hypothesis is that differences in upbringing, which were omitted from the present study, partially explain our findings. Religiosity, as well as cultural orientations, are largely the product of upbringing and remain relatively stable over generations (Y. Chen & VanderWeele, 2018; Kallatsa, Spännäri, & Tervo-Niemelä, 2023). Finally, differential exposure may account for these findings; users in countries with high religiosity levels, such as Nigeria and South Africa (with 69.3% and 57.3%, respectively mention being religious), may be targeted more frequently, an exposure variation that the present study did not fully capture.

From a knowledge perspective, technical expertise was unexpectedly associated with *higher* phishing victimization. This counterintuitive

pattern, also observed in other studies (Desolda, Ferro, Marrella, Catarci, & Costabile, 2021; Parsons et al., 2013), may result from overconfidence among technically skilled users or from increased exposure due to heavier internet use, or a combination of both. In contrast, the ability to recognize legitimate emails—rather than phishing emails—was protective, supporting the view that specific, context-relevant knowledge, such as information on various forms of fraud, is more useful than general digital literacy (Gopavaram et al., 2021; Junger, Koning, Hartel, & Veldkamp, 2023). Recognizing phishing messages, on the other hand, did not reduce victimization, possibly because some participants over-identified messages as fraudulent, reducing discriminative accuracy. Contrary to expectations, privacy orientation was unrelated to victimization, echoing Colnago, Cranor, and Acquisti (2023), who found that self-reported privacy concern does not necessarily translate into preventive behavior.

Consistent with prior research, self-control emerged as one of the strongest predictors of phishing victimization. Individuals reporting lower self-control and higher willingness to engage in risky behaviors, such as sharing login credentials, were significantly more likely to be victimized. This supports theoretical frameworks identifying self-control as a central behavioral regulator (Borghans et al., 2008; Duckworth & Kern, 2011; Langenderfer & Shimp, 2001) and extends empirical findings from recent cybersecurity studies (Halevi, Memon, & Nov, 2015; Vishwanath et al., 2018).

Although Thoughtfully Reflective Decision-Making (TRDM) was related to phishing victimization in the bivariate analysis, this association disappeared in the multilevel analysis, diverging from earlier work linking TRDM to self-protective online behaviors (Howell, 2021; Howell et al., 2024). This suggests that while reflective thinking may facilitate caution in controlled settings, it might not override the habitual, automatic processes that dominate email handling in everyday contexts.

Evidence of repeat victimization also emerged: individuals who had experienced consumer fraud were more likely to fall victim to phishing. This finding reflects well-documented patterns in both offline and online crime, where victimization clusters among individuals with higher exposure or lower self-control (Farrell & Pease, 2018; Moneva, Miró-Llinares, & Hart, 2020; Snyder & Golladay, 2024). Repeated targeting may also result from attackers exploiting previously compromised personal information or contact lists.

Exposure-related factors played a notable role. Participants with more email accounts were significantly more likely to be victimized, consistent with prior evidence linking broader attack surfaces to increased online risk (Edwards et al., 2017; Tajalizadehkhoo et al., 2017). Surprisingly, most email management habits—including the frequency with which people check emails and how they organize them—were unrelated to the outcome of phishing attacks, with the exception of participants who used their inbox as a to-do list, who were less likely to fall victim. This counterintuitive result may reflect heightened engagement with incoming messages. However, the email-management scale may have lacked sufficient sensitivity to capture nuanced behavioral strategies (Ge, Lu, Cui, Chen, & Qu, 2021).

Socio-demographic and economic variables showed minimal impact. Neither age nor gender predicted phishing victimization, consistent with prior findings (Gopavaram et al., 2021; Holtfrete et al., 2008). The limited age range (mostly under 27 years old) likely restricted variability. Similarly, macro-level indicators—Gross Domestic Product (GDP), and ICT Development Index (IDI)—were unrelated to phishing, contrasting with large-scale studies linking national wealth and technological development to cybercrime rates (Dearden & Bergeron, 2026; Kigerl, 2016; Srivastava et al., 2020). This discrepancy may reflect the relative digital homogeneity of university students, who generally possess high internet access regardless of national context.

In examining phishing recognition, participants who correctly identified more phishing messages tended to score lower on masculinity

and risk-taking and were slightly older—traits associated with general risk aversion (Figner & Weber, 2011; Sheaffer, Bogler, & Sarfaty, 2011). These associations were small but consistent with the interpretation that high recognition scores partly reflect cautiousness rather than superior detection skills. Interestingly, having multiple email accounts was linked to both higher recognition and higher victimization, suggesting that greater exposure may simultaneously improve detection ability and increase risk.

Similarly, legitimate email recognition was higher among participants with more email accounts and those who did not restrict checking to specific times, possibly indicating more active or experienced users. Again, effects were small. An isolated result showing lower performance among Canadian participants remains unexplained.

Overall, the findings emphasize that phishing victimization is shaped primarily by individual traits and behavioral patterns rather than by cultural, demographic, or national factors. Self-control, risk-taking, and exposure emerged as the most robust predictors, while cognitive and contextual factors—such as TRDM, privacy orientation, and cultural dimensions—played limited roles. The results suggest that phishing susceptibility arises less from cultural background and more from habitual, automatic processing of emails and impulsive behavior.

4.2. Answers to the research questions

Looking back at our first research question—How are individual cultural orientations related to phishing victimization?—we conclude that, apart from long-term orientation, no other cultural dimensions were significantly associated with phishing victimization. We propose that this limited cultural effect may be explained by processes of global convergence. As Taras et al. (2016) noted, theories of cultural convergence and modernization have received substantial empirical support, suggesting that cross-border personal and organizational interactions increasingly blur cultural distinctions between countries. Among young, digitally connected populations such as university students, this effect may be amplified by the global reach of social media and online communication, which promote shared norms and behaviors across cultures. Interestingly, low masculinity was associated with higher phishing-recognition accuracy but did not reduce actual victimization, suggesting that cultural traits may influence cognitive vigilance without necessarily translating into protective behavior.

Turning to our second research question—How do risk factors other than cultural orientations relate to being a victim of phishing?—our results support a straightforward explanatory model. First, self-control emerged as the strongest predictor of phishing victimization. The ability to regulate impulses and engage in reflective rather than automatic decision-making appears to be the primary determinant of susceptibility. Second, exposure, operationalized as the number of email accounts, was the next most important factor. Individuals who maintain multiple accounts are more likely to encounter phishing attempts and, consequently, face a greater risk of victimization—even though their experience may slightly improve recognition accuracy. Finally, specific, context-relevant knowledge on fraud, reflected in the ability to identify legitimate emails, also played a protective role, though its effect was smaller than those of self-control and exposure.

Our variance analyses revealed that 80–90% of the variation in phishing-related outcomes occurred within locations rather than between them. In other words, participants differed more from one another within countries than across countries. While our explanatory variables effectively accounted for cross-location differences in phishing victimization, they were less successful in explaining variation in phishing and legitimate email recognition. These findings reinforce the conclusion that individual differences—in self-control, exposure, and context-relevant knowledge—play a far more decisive role in phishing vulnerability than broader cultural or contextual factors.

4.3. Implications and recommendations

4.3.1. Measuring phishing: victimization versus scenario-based measures

Our findings reveal minimal overlap ($r=.128$) between the explanatory factors associated with the three outcome variables—phishing victimization, phishing recognition, and legitimate email recognition—indicating that these measures capture distinct constructs rather than different facets of the same phenomenon. This challenges our initial assumption that these three concepts all measured ‘phishing susceptibility’ and suggests that each reflects a unique aspect of user behavior and cognition.

Victimization is the most direct and meaningful indicator of real-world phishing susceptibility. Although victimization surveys have well-known limitations, they remain the most reliable tool for international comparisons of cybercrime prevalence (Aebi et al., 2002; Dijk, Kesteren, & Smit, 2007; Messner & Zimmerman, 2014; National Academies of Sciences et al., 2018).

The scores on the phishing recognition scale, besides highlighting the ability to recognize phishing, also reflect a more general tendency toward cautiousness. The scores on legitimate email recognition seem to capture context-relevant knowledge about email and URL structure, representing the most direct measure of phishing-relevant expertise. While this context-relevant knowledge can reduce victimization risk, its protective effect is modest. As Jones, Towse, and Race (2015) and Kahneman (2011) noted, scenarios as those used in the present study may prompt System 2 thinking—slow, deliberate information processing—rather than measure actual phishing vulnerability.

These distinctions caution against equating scenario-based measures with real-world phishing vulnerability, as done in several prior studies (Anandpara et al., 2007; Canfield et al., 2016; Canfield, Fischhoff, & Davis, 2019; Furnell, 2007; Jerrim, 2023; Parsons et al., 2015). Instead, our findings align with critiques highlighting the conceptual limitations of simulated scenarios (Anandpara et al., 2007; Crgol & Vrhovec, 2024; Parsons et al., 2013, p. 195). Future research should therefore differentiate explicitly between behavioral victimization measures and cognitive scenario-based assessments when evaluating phishing susceptibility.

4.3.2. Policy recommendations

The present findings suggest that, within culturally diverse organizations, individual cultural orientations may play a smaller role in cybersecurity behavior than previously assumed. This does not imply that organizational culture is unimportant, rather, prior research has demonstrated that security-related norms, leadership practices, and collective values can shape employees’ cybersecurity behavior (Huang & Pearson, 2019; Lacey, 2010; Okere, Van Niekerk, & Carroll, 2012). However, such dynamics were beyond the scope of the current study. Of course, based on the current data, we cannot conclude that culture is irrelevant in all phishing contexts. Specific emails with sexual content, for example, could be influenced by cultural orientations. This was not investigated in the present study.

Phishing defenses should not primarily be tailored around national or cultural stereotypes, but should instead focus on behavioral indicators observable at the individual level (impulsivity, risk-taking, high-exposure). The above implies that self-control should be treated as a security variable, meaning that we should design security interventions that reduce time pressure and impulsive action. This has interesting trade-offs with typical usability heuristics, as it may introduce additional friction to ‘force’ the user to pause and reflect on actions, thereby trying to counter momentary lapses in self-regulation. Designing systems that anticipate impulsive behavior may be more impactful than further refining awareness of phishing cues alone.

Based on our results, we offer two primary recommendations for interventions.

First, our findings suggest that, as context-relevant knowledge matters, phishing education matters. While some have questioned the long-

term effectiveness of awareness campaigns (Bada, Hutchings, Papadodimitraki, & Clayton, 2023), the key issue is not whether education works, but how it can be made effective. Public awareness initiatives should focus on context-relevant knowledge such as explaining the modus operandi of fraudsters (Junger et al., 2023) and, more specifically, on teaching users to interpret URLs and sender addresses, employ interactive game-based training, and cultivate a habit of questioning the sender’s identity: ‘Who is the person sending this email?’ (Bartolucci, Aquilino, Bril, Duncan, & van Steen, 2023; Bullee & Junger, 2020; Qawasmeh, AlQahtani, & Khan, 2025; Somestad & Karlzén, 2019). To be clear, we do not state that technical defenses can be replaced by behavioral interventions. The ideal would be that technical solutions would go hand in hand with social solutions to strengthen each other. For instance, technical solutions can help with domain highlighting or a browser plugin what goes over all URLs in the email/page and spots inconsistencies.

Second, given that self-control emerged as a crucial protective factor, training programs should aim to strengthen users’ situational self-control strategies’ (Duckworth, Gendler, & Gross, 2016). One effective approach is the ‘Stop, Think, and Check’ protocol proposed by Nguyen, Jensen, and Day (2021), which encourages users to pause before acting, consider contextual cues, and verify legitimacy—demonstrating positive outcomes in phishing prevention (Nguyen et al., 2021, p. 8). Together, these recommendations highlight that effective anti-phishing policies should combine *contextual, knowledge-based education with behavioral self-regulation strategies*, enabling users not only to recognize threats but also to resist impulsive responses upon encountering them.

4.4. Study limitations

This study relied on data collected from university students. Students are often used as a target population in cyber security research (Frauenstein & Flowerday, 2020) as well as in other fields (Jager, Putnick, & Bornstein, 2017). Convenience samples – and student-based samples in particular – have been criticized, mostly for the limited ability for generalizability to a larger population (Andrade, 2021; Ashraf & Merunka, 2017; Jager et al., 2017). However, student samples also have specific advantages that are relevant for the present research. Beyond being cost-effective, student samples are relatively homogeneous with respect to socio-economic characteristics and more comparable across countries than random population samples would be, for instance regarding age, education, work experience, social class, and their social environment (Ashraf & Merunka, 2017; Highhouse & Gillespie, 2010; Peterson & Merunka, 2014; Winton & Sabol, 2022). Furthermore, students have the advantage that typical organizational influences that might impact their attitudes and/or behavior are limited.

More detailed comparisons of student samples with other samples have been effectuated. (Kees, Berry, Burton, & Sheehan, 2017-author-year) found that reliabilities of important concepts are similar in student samples and other samples and that the data quality of students samples was better than that of commercial samples. Also, effect sizes in student samples are comparable to those obtained from non-student populations (Highhouse & Gillespie, 2010; Wheeler, Shanine, Leon, & Whitman, 2014). Significantly for the present study, Hanel and Vione (2016) analyzed the differences between student and representative samples in terms of their values in the ‘World Values Survey’ (N=84,737 participants, 59 countries). They report that ‘students were on average as heterogeneous as the general public both between and within countries’ (Hanel & Vione, 2016, p. 5). Similar results were reported by Highhouse and Gillespie (2010). Please note that we found considerable differences on the five cultural orientations scales by ‘country in which the respondent grew up’ (see (section 3.1)). These differences are in line with Hanel and Vione (2016) and Highhouse and Gillespie (2010)’s results. In contrast with these findings, however, Taras, Kirkman, and Steel (2010) found that cultural effects tend to be more pronounced in older samples. Finally, students are at a relatively

high risk: younger individuals generally spend relatively more time online (ITU, 2024); they are more likely to be exposed to phishing attempts (Koning et al., 2023) and to become victims of phishing than older users (Nurse, Karppinen, Milward, & Varughese, 2023; Office for National Statistics, 2022). It needs to be emphasized that, while student samples have benefits, the overall generalizability of our findings to broader populations, such as working professionals and older users, should not be taken for granted and still needs to be investigated.

Our study relies solely on self-reported surveys and a scenario-based phishing task. We did not have access to more objective data. In this sense, our analysis remains largely descriptive. *Nevertheless, self-reports of victimization remain the best measure of crime, especially in a cross-national context* (National Academies of Sciences et al., 2018), as mentioned above. Self-reports of values (Fisher & Katz, 2000), personality (Paulhus & Vazire, 2007), and of sensitive questions (Corneille & Gawronski, 2024) also have reasonable to good validity, particularly in self-administered surveys (Tourangeau & McNealey, 2003; Tourangeau & Yan, 2007). Therefore, we believe that our study nonetheless provides relevant insights on the potential phishing susceptibility based upon the views expressed, and offers a baseline from which further practical phishing assessments could now be tailored.

Another limitation concerns the level of analysis: individual-level data were combined with country-level indicators of economic and technological development. Because these operate at different analytical levels, interpretations involving such variables should be treated with caution.

Finally, this study did not examine the content or design of phishing messages, which can strongly influence victimization (Sommestad & Karlzén, 2019). With advances in artificial intelligence, phishing messages are likely to become increasingly sophisticated (Caldwell, Andrews, Tanay, & Griffin, 2020; Goodman, 2015; Olber, 2023). In addition, the cross-sectional design precludes causal inference; the observed associations identify relationships but not their direction.

5. Future research

Given the limitations of this study, future research should seek to replicate and extend our findings using culturally diverse and demographically broader samples. Furthermore, we did not measure acculturation indicators such as migration history, years lived in the current country, or language use, which limits our ability to disentangle primary cultural socialization from ongoing cross-cultural exposure. Future research should incorporate explicit measures of acculturation and examine more demographically diverse samples. As global workforces become increasingly heterogeneous, understanding how cultural and individual differences interact to shape cybersecurity behavior remains an important and timely research priority. The present study represents an initial step in this direction.

Future work should also examine different facets of phishing in greater depth. The email scenarios employed here may vary in relevance or interpretability across participants and cultural contexts. Phishing recognition depends not only on comprehension of the presented content but also on individuals' sense-making and contextual alignment with the message. To capture these complexities, subsequent studies could combine scenario-based recognition tasks with complementary assessments—such as knowledge-based quizzes, behavioral simulations, or field experiments using controlled phishing campaigns—to obtain a more comprehensive understanding of user susceptibility.

6. Conclusions

This study contributes to the growing body of cross-cultural cybersecurity research, addressing the imbalance in existing literature that remains heavily focused on Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies (Henrich et al., 2010; Kaur et al., 2021).

Our findings indicate that phishing victimization is influenced not only by situational factors such as exposure and knowledge, but also by a key individual characteristic—low self-control. Although knowledge and exposure clearly play a role, the data suggest that many participants process their emails rapidly, relying primarily on automatic, intuitive 'System 1' thinking rather than deliberate reasoning. This tendency underscores why self-control emerges as a particularly strong predictor of actual victimization.

These patterns were consistent across the 14 universities in our study that spanned four continents. Cultural orientations, in contrast, had minimal influence on phishing susceptibility. While our explanatory variables accounted well for phishing victimization, predicting the recognition of both phishing and legitimate emails proved more challenging, and differences between locations remained largely unexplained. We propose that participants' interpretations of the email scenarios may partly reflect different underlying mechanisms—cautious behavior driving phishing recognition and knowledge driving legitimate email recognition.

Overall, our findings suggest that within culturally diverse organizations, cultural factors may play a smaller role in cybersecurity vulnerabilities than previously assumed. Instead, fostering deliberate, reflective decision-making and supporting users' self-control and attention may be more effective in reducing phishing risks across cultural contexts.

CRedit authorship contribution statement

Marianne Junger: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Paweł Olber:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Rafał Płocki:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **J.W. (Hans) Luyten:** Writing – review & editing, Methodology, Formal analysis. **Luka Koning:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Caitlyn N. Muniz:** Writing – review & editing, Investigation. **J.H. (Jan-Willem) Bullee:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Victoria Wang:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Reinhardt Botha:** Writing – review & editing, Methodology, Investigation, Conceptualization. **C. Jordan Howell:** Writing – review & editing, Data curation. **Verena Distler:** Writing – review & editing, Investigation. **Xiao Wei Chen:** Writing – review & editing, Data curation. **Hiep Cong Pham:** Writing – review & editing, Investigation. **Mohammed Aljohani:** Validation, Resources, Data curation, Conceptualization. **Newman U. Richards:** Writing – review & editing, Investigation, Data curation. **Fabian Muhly:** Writing – review & editing, Investigation. **Abhishta Abhishta:** Writing – review & editing, Investigation. **Steven Furnell:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no financial interests or personal relationships that could have influenced this research.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cose.2026.105050](https://doi.org/10.1016/j.cose.2026.105050).

Appendix A

Table A1

Locations for recruiting participants classified by location, frequencies, percentage, and response rates.

Locations	Participants who completed the questionnaire		Participants, who started		Response
	Freq.	Percent	Freq.	Percent	
Africa					
1. Nigeria, University of Nigeria	155	7.1%	310	11.1%	50%
2. South Africa, Nelson Mandela University	78	3.6%	194	7.0%	40%
Asia					
3. India, Indian exchange students with the University of Twente (Netherlands) [1]	29	1.30%	35	1.30%	83%
4. Vietnam, University of Economics HCMC (UEH)	196	8.9%	264	9.5%	74%
Europe					
5. Luxembourg, University of Luxembourg	6	0.3%	15	0.5%	40%
6. Germany, LMU Munich [2]	77	3.5%	92	3.3%	84%
7. Netherlands, University of Twente	161	7.30%	176	6.30%	91%
8. Poland, Police Academy in Szczytno	308	14.0%	319	11.4%	97%
9. Switzerland, ETH Zurich	59	2.7%	89	3.2%	66%
10. United Kingdom, University of Nottingham	114	5.2%	149	5.3%	77%
11. United Kingdom, University of Portsmouth	72	3.3%	104	3.7%	69%
North America					
12. Canada: Simon Fraser University	281	12.8%	357	12.8%	79%
13. United States, University of Texas El Paso	478	21.8%	486	17.4%	98%
14. United States, University of South Florida	179	8.2%	200	7.2%	90%
	2193	100%	2790	100%	79%

[1] These students stayed in the Netherlands for 1 month.

[2] 36 German students were recruited through Prolific academic.

Table A2

Descriptive information of the variables in the analysis.

	Missing	Mean	Median	SD	Skew-ness	Min.	Max.
Recognition phishing emails	0	3.9	4.0	1.52	-.24	0	6
Recognition legitimate emails	0	3.2	3.0	1.45	-.03	0	6
Phishing victim	0	.1	.0	.28	3.03	0	1
Power distance	0	-1.1	-1.0	.62	.84	-2	2
Uncertainty avoidance	0	1.0	1.0	.64	-1.04	-2	2
Collectivism	0	.0	.0	.66	.00	-2	2
Long-term orientation	0	1.0	1.0	.59	-1.49	-2	2
Masculinity	0	-.7	-.8	.94	.38	-2	2
Do you consider yourself a practicing religious person?	0	1.2	1.0	.62	-.16	.0	2.0
Nationality, in which country did you mainly spend your childhood? ¹	See online attachment						
Privacy orientation	0	.3	.5	.91	-.29	-2	2
Self-control	0	.4	.5	.39	-.65	-.92	1
TRDM	0	.8	1.0	.79	-.85	-2	2
Technical training	2	-.1	-.3	.36	1.68	-.30	.72
Victim online consumer fraud	0	.1	.0	.32	2.44	0	1
Risk-taking	0	1.1	1.0	.96	.44	0	3
Number of email accounts	0	2.5	2.0	7.51	31.71	0	300
How many emails do you receive on average per day?	0	1.4	1.0	.79	2.79	1	6
Organized & structured email management	0	2.5	2.0	1.01	.37	1	5
I restrict myself to checking my email at specific times of the day	0	2.1	2.0	1.15	.74	1	5
I keep email messages in my inbox as a reminder of things I need to do	0	3.0	3.0	1.32	-.18	1	5
I leave emails messages in the inbox after I have read them	0	3.7	4.0	1.18	-.74	1	5
Age N=2094 ¹	49	22.2	21.0	4.80	2.38	17	40
Gender ¹	53	.35	.0	.48	.65	0	1
GDP	0	88.9	93.6	13.03	-2.50	46.9	96.7
IDI	0	48,500.7	53,372	30,924.7	-.20	1621	128,259

¹ Missing values were coded as a separate category in the multilevel analysis.

Data availability

Data will be made available on request.

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